

Jack H. Madden

RESEARCH ARTIST @ BLUE MARBLE SPACE INSTITUTE OF SCIENCE

PH.D. IN ASTROPHYSICS FROM CORNELL, M.F.A. IN DIGITAL+MEDIA FROM RISD (HE/HIM/HIS)

@Astro_Madden | jack.madden@bmsis.org | jackhadden.github.io | JackHMadden | ORCID 0000-0002-4701-7833

Astrophysicist and artist working on humanizing our understanding of the universe through research-based creation. Blending research backgrounds in remote sensing, exoplanet climate modeling, and astronomy education with experience in new media art, creative computing, data visualization, and research-creation to address problems in astronomy from a new perspective.

Recent Highlights

- 2026 Lunar Sample Return (LSR-1), Sculpture using lunar material being sent to the Moon in June. [link](#)
- 2026 Art and Intelligence, Book chapter in prep with Penn State Rock Ethics Institute
- 2025 Research Affiliate, Blue Marble Space Institute of Science [link](#)
- 2023 TEDxRISD, How to make Art like an Astrophysicist [link](#)

Education

M.F.A. Rhode Island School of Design - Thesis: Abyss without Vertigo

[Providence, Rhode Island](#)

DIGITAL+MEDIA - ADVISED BY SHONA KITCHEN

[Sept. 2020 - May 2022](#)

Ph.D. Cornell University - Thesis: The Color of Habitability

[Ithaca, New York](#)

ASTROPHYSICS - M.S. AWARDED IN 2017 - ADVISED BY DR. LISA KALTENEGGER

[Sept. 2014 - June 2020](#)

B.A. Franklin and Marshall College

[Lancaster, Pennsylvania](#)

ASTROPHYSICS - ADVISED BY DR. FRONEY CRAWFORD III

[Sept. 2010 - May 2014](#)

Peer Reviewed Papers

- 2024 N. Kutsop, A. G. Hayes, **et al.**, The history and processes of Titan's equator from the geospatial-topology of spectrally distinct units ([ADS](#)) [Icarus](#)
- 2021 L. Coelho, **J. Madden**, L. Kaltenegger, S. Zinder, W. Philpot, M. G. Esquivel, J. Canário, R. Costa, W. Vincent, Z. Martins, Color catalogue of life in ice: Surface biosignatures on icy worlds ([ADS](#)) [Astrobiology](#)
- 2020 **J. Madden**, & L. Kaltenegger, High-resolution Spectra for a Wide Range of Habitable Zone Planets around Sun-like Stars ([ADS](#)) [ApJL](#)
- 2020 **J. Madden**, & L. Kaltenegger, How surfaces shape the climate of habitable exoplanets ([ADS](#)) [MNRAS](#)
- 2020 L. Kaltenegger, Z. Lin, & **J. Madden**, High-Resolution Transmission Spectra of Earth through Geological Time ([ADS](#)) [ApJL](#)
- 2020 **J. H. Madden**, S. Pandita, B. Kim, J. P. Schuldt, A. S. Won & N. G. Holmes, Ready Student One: Exploring predictors for student learning in virtual reality ([ADS](#)) [PLOS ONE](#)
- 2019 L. Kaltenegger, **J. Madden**, Z. Lin, S. Rugheimer, A. Segura, R. Luque, E. Pallé, N. Espinoza, The Habitability of GJ 357 d: Possible Climates and Observability ([ADS](#)) [ApJL](#)
- 2019 R. Luque **et al.**, Planetary system around the nearby M dwarf GJ 357 including a transiting, hot, Earth-sized planet optimal for atmospheric characterization ([ADS](#)) [A&A](#)
- 2018 **J. Madden**, & L. Kaltenegger, A Catalog of Spectra, Albedos, and Colors of Solar System Bodies for Exoplanet Comparison ([ADS](#)) [Astrobiology](#)
- 2018 **J. H. Madden**, A. S. Won, J. P. Schuldt, B. Kim, S. Pandita, Y. Sun, T. J. Stone, & N. G. Holmes, Virtual Reality as a Teaching Tool for Moon Phases and Beyond ([ADS](#)) [PERC Proceedings](#)
- 2014 C. Neish, **J. Madden**, L. Carter, B. Hawke, T. Giguere, V. Bray, G. Osinski, & J. Cahill, Global Distribution of Lunar Impact Melt Flows ([ADS](#)) [Icarus](#)
- 2013 J. Ridley, F. Crawford, D. Lorimer, S. Bailey, **J. Madden**, R. Anella, & J. Chennamangalam, Eight New Radio Pulsars in the Large Magellanic Cloud ([ADS](#)) [MNRAS](#)

Art Exhibitions

SOLO AND GROUP

2021	[SOLO] Curator: Deborah Clemons - Dorner Prize (Complete Definitions)	RISD Museum
2026	Griffin-1 Lunar Mission (Lunar Sample Return (LSR-1))	MoonGallery
2025	ColorShifts (Spectrophotographic Photos)	ISCC
2024	Inside Stars and Bodies (Poster Session, 29 Atmospheres)	AbSciCon
2024	UnFiguring (Known Displeasures)	Kirkland Gallery
2023	Some Tumblrs! (SlurpeeBlog)	Rhizome
2023	Miniature Monumental (22 Atmospheres, The Individual)	Bristol Art Museum
2023	The Art of Planetary Science (22 Atmospheres, The Otherview Effect)	University of Arizona
2022	Grad Thesis Show (Untitled Space no.4)	RISD
2022	2nd Festival of the Smallest (The Individual)	222Lodge
2022	Transitory Void (see Equations)	Boston CyberArts
2022	1+1=22 (see Equations)	Sol Koffler Gallery
2021	NG-17 test flight to International Space Station (The Individual)	MoonGallery
2020	Pandemic Publishing (Orthodox Nihilism)	volume.1
2020	Code as Medium (Books for Robots (only))	Places Instead
2020	Alone/Together (Untitled)	IncuArts Gallery

In Media

5.27.22	Art and design on display at the 2022 RISD Graduate Exhibition , Kris Craig	Providence Journal
12.13.21	Astrophysicist Earns Dorner Prize , Simone Solondz	RISD News
11.1.20	Bringing Exoplanets to Life , Christian Fogerty	StarDate Magazine
10.25.20	The Color of Habitable Worlds , Matthew Cimone	Universe Today
8.8.20	Discussed: What If We Lived on a Super Earth? - with Jack Madden , What If	YouTube
5.23.20	New Planetary Color Models Will Decode Signs Of Extrasolar Life , Bruce Dorminey	Forbes
5.18.20	Astronomers develop 'decoder' to gauge exoplanet climate , Blaine Friedlander	Cornell Chronicle
3.25.20	Video game experience or gender may improve VR learning, study finds , Melanie Lefkowitz	Cornell Chronicle
10.7.19	Leading Lines Podcast Episode 65: Jack Madden and Swati Pandita , Derek Bruff	Leading Lines
7.31.19	TESS satellite uncovers 'first nearby super-Earth' , Blaine Friedlander	Cornell Chronicle
2.5.19	Study probes effect of virtual reality on learning , Linda Glaser	Cornell Chronicle
9.19.18	One (Solar System) catalog to aid them all , Amber Hornsby	Astrobites.org
7.31.18	This Solar System Catalog Could Be Key to Finding an Earth-Like Exoplanet , Ryan Mandelbaum	Gizmodo.com
7.26.18	Exoplanet detectives create catalog of 'light-fingerprints' , Linda Glaser	Cornell Chronicle
3.14.18	Elevator Art Contest Winners , Melanie Lefkowitz	Cornell Library
9.13.12	F&M Student Discovers Rare Extragalactic Pulsar , Chris Karlesky	F&M News
10.23.12	F&M student makes rare scientific discovery , Jere Gish	WGAL 8 TV

Awards, Fellowships, & Residencies

ART

2025	Best in Show - Professional	ISCC ColorShifts
2023	Miniature Monumental Recognition award	Bristol Art Museum
2022	Get Visual Award	Wolfram
2021	RISD Museum Dorner Prize	RISD
2021	Artist Residency at Wendy.Network	Virtual
2021	Nature Lab Vis-a-thon Collaborator	RISD
2020	RISD Tuition Fellowship	RISD

SCIENCE

2019	Brinson Foundation research funding	<i>Cornell</i>
2018	Branson and Edna B. Shelley Service Award	<i>Cornell</i>
2017	Center for Teaching Innovation Graduate Research Teaching Fellowship	<i>Cornell</i>
2016	Branson and Edna B. Shelley Outstanding Teaching Assistant Award	<i>Cornell</i>
2016	NY Space Grant Fellowship	<i>Cornell</i>
2014	Honors Societies: Phi Beta Kappa, Sigma Xi, Sigma Pi Sigma	<i>F&M</i>
2013	Kershner Scholar	<i>F&M</i>
2013	Micheal J. Mumma Prize in Physics and Astronomy	<i>F&M</i>
2012	Hackman Summer Research Scholarship	<i>F&M</i>

Essays

2026	J. Madden , Art and Intelligence (Chapter for Alien Horizons - In Prep)	<i>Rock Ethics Institute</i>
2026	J. Madden , The Constructed Beauty of the Universe (Accepted)	<i>Kosmica Institute</i>
2024	J. Madden & J. Roberts , Double Take: Plate VI Maskeyne Da Region of the Moon (Issue 19)	<i>RISD Manual</i>

My Courses

2023	Introduction to Computational Art , Undergraduate	<i>RISD</i>
2022	Digital+Media Studio/Seminar , Graduate	<i>RISD</i>
2022	Design Studio , Undergraduate	<i>RISD</i>
2022	Astrophysics for Artists , Undergraduate/Graduate	<i>RISD</i>

Guest Lectures and Public Talks

2026	Envisioning Join Futures in Art & Astronomy , PALLAS	<i>AAS</i>
2026	Research Creation: Art and Science Collaboration in Astronomy , Franklin & Marshall College	<i>Lancaster, PA</i>
2025	Beyond the Eye: Making of the Modern Astronomical Image , Undergraduate Art History Society	<i>Harvard</i>
2024	A future for art in astrophysics research , CCC Lab	<i>Bamberg, Germany</i>
2023	How to make art like an astrophysicist , TEDxRISD	<i>Providence, RI</i>
2022	Light Pollution , DM-7152 RESEARCH STUDIO: TECHLANDS	<i>RISD</i>
2022	A guide to the anthro-post-centric universe , DM-1551 SPECULATIVE SPECIES	<i>RISD</i>
2022	Theoretical Photorealism , DM-1560 DEEPFAKES	<i>RISD</i>
2021	Frontier Science Visualizations , DM-1519 LITERACY_IN_3D.OBJ	<i>RISD</i>
2019	How we see the sky , ASTRO1101 Introductory Astronomy	<i>Cornell</i>
2018	Searching for Intelligent Life in Cornell Classrooms and Beyond , Fuertes Observatory	<i>Ithaca, NY</i>
2018	The New Search for Life , Tompkins County Public Library	<i>Ithaca, NY</i>
2018	How We Search for Life on Other Planets , Museum of The Earth, Darwin Days	<i>Ithaca, NY</i>
2017	Causality and Black Holes , ASTRO1101 Introductory Astronomy	<i>Cornell</i>
2015	Black Holes , ASTRO1101 Introductory Astronomy	<i>Cornell</i>
2015	Are we alone? , Mann Library, SPARK Talk	<i>Ithaca, NY</i>

Science Research Experience

Cornell Astronomy and Space Sciences

Ithaca, NY

GRADUATE RESEARCH ASSISTANT - DR. LISA KALTENEGGER

Fall 2014 - Summer 2020

- Calculated a catalog of spectra and albedos for Solar System objects as references in exoplanet characterization.
- Updated and optimized 1D climate and photochemistry models, and observation simulations for exoplanet use.
- Modeling of the climate and photochemistry of terrestrial exoplanets to determine suitable conditions for life and detectable biosignatures in regard to the effect of surface albedo.
- Modeled the climate and determined the habitability of the planet Gl 357 d.
- Created a database of habitable exoplanet models and high resolution observations for different surfaces types.

Cornell Physics Education Research Lab

Ithaca, NY

GRADUATE RESEARCH ASSISTANT - DR. NATASHA HOLMES

Fall 2018 - Spring 2019

- Explored the differences in learning outcomes between virtual reality, computer simulation, and hands-on activities for Moon phases.
- Investigated demographic links to learning outcomes by condition.
- Designed and built a full Moon phase demonstration using the Unity game engine for Oculus Rift.

Goddard Spaceflight Center

Greenbelt, MD

SUMMER INTERNSHIP PROGRAM - DR. LYNN CARTER & DR. CATHERINE NEISH

Summer 2013

- Scanned the entire Moon for lunar impact melts and cataloged their features.
- Discovered 24 new impact melts and updated the global melt statistics.

Franklin and Marshall College

Lancaster, PA

UNDERGRADUATE RESEARCH ASSISTANT - DR. FRONEY CRAWFORD III

Fall 2010 - May 2014

- Investigated pulsar candidates in the Small and Large Magellanic clouds using data from the Parkes Multibeam Pulsar Survey and tested image recognition techniques for pulsar identification.
- Discovered PSR J0456-69, one of only 28 known extragalactic pulsars at the time.

Teaching Experience

Lecturer

Providence, RI

RHODE ISLAND SCHOOL OF DESIGN

- Fall 2022: Introduction to Design
- Fall 2022: Digital+Media MFA Studio/Seminar
- Winter 2023: Introduction to Computation
- Winter 2022: Astrophysics for Artists

Certificate in Collegiate Teaching in Art and Design

Providence, RI

RHODE ISLAND SCHOOL OF DESIGN

Fall 2021 - Winter 2022

- Course I designed was competitively selected to be taught in RISD's curriculum.
- 2 semesters of collegiate teaching and practicum.

Graduate Research Teaching Fellow

Ithaca, NY

CORNELL UNIVERSITY

Fall 2017 - Spring 2018

- 2 semesters of pedagogy and teaching as research courses.
- Conducted original education research on VR for physics labs.
- Designed and taught 4 workshops for graduate students on teaching and course management.

Head Teaching assistant

Ithaca, NY

CORNELL UNIVERSITY

Spring 2016

- Head teaching assistant for 1 semester. Extensive course management and leading of TA activities.
- Designed and taught 2 discussion sections per week.
- Worked with faculty to revamp the policies and procedures for TAs and Head TAs.
- Created an online archive of course material and guides for TAs.

Teaching Assistant

Ithaca, NY

CORNELL UNIVERSITY

Fall 2014 - Fall 2015

- 3 semesters of designing and teaching 2 discussion sections per week.
Wrote lesson plans, created homeworks, held office hours, review sessions, and graded.

Undergraduate Teaching Assistant

Lancaster, PA

FRANKLIN AND MARSHALL COLLEGE

Fall 2013 - Spring 2014

- Teaching assistant for 2 semesters.
Helped with lectures, wrote assignments, held office hours, and graded.

Tutor and lab instructor

Lancaster, PA

FRANKLIN AND MARSHALL COLLEGE

Fall 2010 - May 2014

- Astronomy and physics tutor for 4 years. Covered 1st and 2nd semester physics, astrophysics, and astronomy.
- Lab assistant for 1st, 2nd, and 3rd semester physics, and observational astronomy.

Conference Talks

AbSciCon 2026

FUTURES OF ASTROBIOLOGY: SEARCHING FOR LIFE ALONGSIDE MILITARY CONTRACTORS

[Madison, WI](#)

May 2026

AbSciCon 2024

COLLABORATING WITH ARTISTS IN THE SEARCH FOR LIFE

[Providence, RI](#)

May 2024

UnFiguring

KNOWN DISPLEASURES

[Harvard University](#)

March 2024

SETI Symposium

THE POWER OF ART IN THE SEARCH FOR LIFE

[Penn State](#)

June 2023

AAS 235

REVEALING THE IMPORTANCE OF SURFACE COLOR IN MODELING HABITABLE EXOPLANET ATMOSPHERES

[Honolulu, HI](#)

January 2020

AAS 235

READY STUDENT ONE: EXPLORING THE PREDICTORS OF STUDENT LEARNING IN VIRTUAL REALITY

[Honolulu, HI](#)

January 2020

AbGradCon

1D EXOPLANET HABITABILITY: NOW IN TECHNICOLOR

[University of Utah](#)

July 2019

ERES V Symposium

EFFECT OF SURFACE TYPE FOR EARTH-LIKE PLANETS ORBITING FGKM STARS

[Cornell University](#)

June 2019

Breakthrough Starshot Workshop

CHIPSAT SCIENCE CASES FOR VENUS AND TITAN

[Auckland, NZ](#)

March 2019

Connecting Teaching and Research Conference

VIRTUAL REALITY AS A TEACHING TOOL FOR MOON PHASES AND BEYOND

[Cornell University](#)

May 2018

ERES IV Symposium

SOLAR SYSTEM BODIES FOR EXOPLANET COMPARISON

[Penn State University](#)

June 2018

American Association of Physics Teachers

VIRTUAL REALITY AS A TEACHING TOOL FOR MOON PHASES AND BEYOND

[Washington D.C.](#)

July 2018

Central Pennsylvania Consortium

IMAGE RECOGNITION TO FIND PULSARS

[Lancaster, PA](#)

April 2014

Professional Service

SEI Committee

ASSISTED WITH DIGITAL+MEDIA DEPARTMENT SOCIAL EQUITY AND INCLUSION INITIATIVES.

[RISD](#)

2021

Co-chair - Cornell Astronomy Department Climate and Diversity Committee

FOUNDING MEMBER - COORDINATED TASKS SUCH AS A CREATING A VALUES STATEMENT, TRAININGS, AND METRICS.

[Cornell](#)

2019-2020

ERES V Conference LOC/SOC

SELECTED TALKS, SCHEDULED, AND DESIGNED PRINT MEDIA FOR A SCIENCE CONFERENCE.

[Cornell](#)

2019

President - Astronomy Graduate Network

COORDINATED SEMINARS, SPEAKERS, EVENTS, AND SOCIAL PROGRAMING FOR THE ASTRONOMY GRADUATES.

[Cornell](#)

2017-2018

Scientific Visualizations

CREATED PRESS RELEASE IMAGERY, JOURNAL COVERS, AND RAN SCIENTIFIC GRAPHIC DESIGN WORKSHOPS.

[Cornell/CSI](#)

2016-2021

Emergency Medical Technician - Basic

VOLUNTEER ON CAMPUS AND IN THE COMMUNITY AS AN EMT. APPROX. 3000 HOURS SINCE 2011

[NY, and PA](#)

2011-2020